

```
In [3]: import pandas as pd
import numpy as np
```

```
In [5]: df=pd.read_csv(r"C:\Users\HP\Downloads\creditcard.csv")
df
```

```
Out[5]:
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V2
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	0.098698	0.363787	...	-0.01830
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	0.085102	-0.255425	...	-0.22577
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	0.247676	-1.514654	...	0.24799
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	0.377436	-1.387024	...	-0.10830
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	-0.270533	0.817739	...	-0.00943
...	...	...	...	...	...	...	...	...	...	...	...	...
284802	172786.0	-11.881118	10.071785	-9.834783	-2.066656	-5.364473	-2.606837	-4.918215	7.305334	1.914428	...	0.21345
284803	172787.0	-0.732789	-0.055080	2.035030	-0.738589	0.868229	1.058415	0.024330	0.294869	0.584800	...	0.21420
284804	172788.0	1.919565	-0.301254	-3.249640	-0.557828	2.630515	3.031260	-0.296827	0.708417	0.432454	...	0.23204
284805	172788.0	-0.240440	0.530483	0.702510	0.689799	-0.377961	0.623708	-0.686180	0.679145	0.392087	...	0.26524
284806	172792.0	-0.533413	-0.189733	0.703337	-0.506271	-0.012546	-0.649617	1.577006	-0.414650	0.486180	...	0.26105

284807 rows × 31 columns



```
In [7]: df.isnull().sum()
```

```
Out[7]: Time      0
        V1        0
        V2        0
        V3        0
        V4        0
        V5        0
        V6        0
        V7        0
        V8        0
        V9        0
        V10       0
        V11       0
        V12       0
        V13       0
        V14       0
        V15       0
        V16       0
        V17       0
        V18       0
        V19       0
        V20       0
        V21       0
        V22       0
        V23       0
        V24       0
        V25       0
        V26       0
        V27       0
        V28       0
        Amount    0
        Class     0
        dtype: int64
```

```
In [15]: df = df.drop_duplicates()
```

```
In [17]: df.duplicated().sum()
```

```
Out[17]: 0
```

```
In [19]: df
```

Out[19]:

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V10
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	0.098698	0.363787	...	-0.01830
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	0.085102	-0.255425	...	-0.22577
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	0.247676	-1.514654	...	0.24799
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	0.377436	-1.387024	...	-0.10830
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	-0.270533	0.817739	...	-0.00943
...	...	...	...	...	...	...	...	...	...	...	...	...
284802	172786.0	-11.881118	10.071785	-9.834783	-2.066656	-5.364473	-2.606837	-4.918215	7.305334	1.914428	...	0.21349
284803	172787.0	-0.732789	-0.055080	2.035030	-0.738589	0.868229	1.058415	0.024330	0.294869	0.584800	...	0.21420
284804	172788.0	1.919565	-0.301254	-3.249640	-0.557828	2.630515	3.031260	-0.296827	0.708417	0.432454	...	0.23204
284805	172788.0	-0.240440	0.530483	0.702510	0.689799	-0.377961	0.623708	-0.686180	0.679145	0.392087	...	0.26524
284806	172792.0	-0.533413	-0.189733	0.703337	-0.506271	-0.012546	-0.649617	1.577006	-0.414650	0.486180	...	0.26109

283726 rows × 31 columns



```
In [23]: x=df.drop("Class",axis=1)
         y=df["Class"]
```

```
In [29]: from sklearn.model_selection import train_test_split
         from sklearn.tree import DecisionTreeClassifier
         from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

         x_train, x_test, y_train, y_test = train_test_split(
             x, y,
             test_size=0.2,
             random_state=42
         )
```

```
In [31]: model=DecisionTreeClassifier(random_state=42)
```

```
In [33]: model.fit(x_train,y_train)
```

```
Out[33]:
```

DecisionTreeClassifier

DecisionTreeClassifier(random_state=42)

```
In [35]: y_pred=model.predict(x_test)

acc = accuracy_score(y_test, y_pred)
print("Accuracy Score:", acc)

cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
```

Accuracy Score: 0.9989779015260988

Confusion Matrix:

```
[[56624  32]
 [  26   64]]
```